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Project Title: SQL Mini Project – SmartMart Retail

Introduction

In this project, I worked as a data analyst for SmartMart Retail company.
The company sells products like Electronics, Clothing, and Home Appliances.
The main goal of this project is to analyze:
Best selling products
Sales by cities
Customer behavior
Category performance

Level 1 – Basic SELECT

SELECT * FROM customers;

customer_id	customer_name	city	age
1	Rahul Sharma	Delhi	28
2	Priya Verma	Mumbai	32
3	Amit Singh	Bangalore	25
4	Neha Kapoor	Delhi	29
5	Rohit Mehta	Pune	35
6	Anjali Gupta	Mumbai	27
7	Karan Patel	Ahmedabad	31
8	Sneha Reddy	Hyderabad	26
NULL	NULL	NULL	NULL

SELECT customer_name, city FROM customers;

Result Grid | Filter Rows:

customer_name	city
Rahul Sharma	Delhi
Priya Verma	Mumbai
Amit Singh	Bangalore
Neha Kapoor	Delhi
Rohit Mehta	Pune
Anjali Gupta	Mumbai
Karan Patel	Ahmedabad
Sneha Reddy	Hyderabad

customers 9

SELECT * FROM products WHERE category='Electronics';

product_id	product_name	category	price
101	Laptop	Electronics	80000.00
102	Smartphone	Electronics	40000.00
103	Headphones	Electronics	2000.00
NULL	NULL	NULL	NULL

SELECT * FROM products WHERE price > 10000;

product_id	product_name	category	price
101	Laptop	Electronics	80000.00
102	Smartphone	Electronics	40000.00
107	Refrigerator	Home Appliances	30000.00
108	Air Conditioner	Home Appliances	45000.00
NULL	NULL	NULL	NULL

SELECT DISTINCT category FROM products;

Result Grid | Filter Rows: | Export: | Wrap C

category
Electronics
Clothing
Home Appliances

Level 2 – WHERE Conditions

SELECT * FROM customers WHERE city='Delhi';

customer_id	customer_name	city	age
1	Rahul Sharma	Delhi	28
4	Neha Kapoor	Delhi	29
NULL	NULL	NULL	NULL

SELECT * FROM customers WHERE age > 30;

customer_id	customer_name	city	age
2	Priya Verma	Mumbai	32
5	Rohit Mehta	Pune	35
7	Karan Patel	Ahmedabad	31
NULL	NULL	NULL	NULL

SELECT * FROM products WHERE price BETWEEN 2000 AND 50000;

product_id	product_name	category	price
102	Smartphone	Electronics	40000.00
103	Headphones	Electronics	2000.00
105	Jeans	Clothing	2000.00
106	Microwave	Home Appliances	7000.00
107	Refrigerator	Home Appliances	30000.00
108	Air Conditioner	Home Appliances	45000.00
NULL	NULL	NULL	NULL

SELECT * FROM sales WHERE order_date > '2024-01-12';

order_id	customer_id	product_id	quantity	order_date
1005	5	103	4	2024-01-13
1006	6	106	1	2024-01-14
1007	7	108	1	2024-01-14
1008	8	102	1	2024-01-15
1009	1	103	2	2024-01-16
1010	2	104	5	2024-01-17
1011	3	105	1	2024-01-18
1012	4	101	1	2024-01-19
NULL	NULL	NULL	NULL	NULL

SELECT * FROM sales WHERE quantity > 2;

order_id	customer_id	product_id	quantity	order_date
1003	3	104	3	2024-01-12
1005	5	103	4	2024-01-13
1010	2	104	5	2024-01-17
NULL	NULL	NULL	NULL	NULL

Level 3 – Sorting & Limiting

SELECT * FROM products ORDER BY price DESC;

product_id	product_name	category	price
101	Laptop	Electronics	80000.00
108	Air Conditioner	Home Appliances	45000.00
102	Smartphone	Electronics	40000.00
107	Refrigerator	Home Appliances	30000.00
106	Microwave	Home Appliances	7000.00
103	Headphones	Electronics	2000.00
105	Jeans	Clothing	2000.00
104	T-Shirt	Clothing	800.00
NULL	NULL	NULL	NULL

SELECT * FROM products ORDER BY price DESC LIMIT 3;

product_id	product_name	category	price
101	Laptop	Electronics	80000.00
108	Air Conditioner	Home Appliances	45000.00
102	Smartphone	Electronics	40000.00
NULL	NULL	NULL	NULL

SELECT * FROM customers ORDER BY age ASC;

	customer_id	customer_name	city	age
▶	3	Amit Singh	Bangalore	25
	8	Sneha Reddy	Hyderabad	26
	6	Anjali Gupta	Mumbai	27
	1	Rahul Sharma	Delhi	28
	4	Neha Kapoor	Delhi	29
	7	Karan Patel	Ahmedabad	31
	2	Priya Verma	Mumbai	32
	5	Rohit Mehta	Pune	35
*	NULL	NULL	NULL	NULL

SELECT * FROM sales ORDER BY order_date DESC LIMIT 5;

	order_id	customer_id	product_id	quantity	order_date
▶	1012	4	101	1	2024-01-19
	1011	3	105	1	2024-01-18
	1010	2	104	5	2024-01-17
	1009	1	103	2	2024-01-16
	1008	8	102	1	2024-01-15
*	NULL	NULL	NULL	NULL	NULL

Level 4 – NULL Handling

UPDATE customers
SET city = NULL
WHERE customer_id = 8;

SELECT * FROM customers WHERE city IS NULL;

Result Grid | Filter Rows:

	customer_id	customer_name	city	age
▶	8	Sneha Reddy	NULL	26
*	NULL	NULL	NULL	NULL

SELECT * FROM customers WHERE city IS NOT NULL;

customer_id	customer_name	city	age
1	Rahul Sharma	Delhi	28
2	Priya Verma	Mumbai	32
3	Amit Singh	Bangalore	25
4	Neha Kapoor	Delhi	29
5	Rohit Mehta	Pune	35
6	Anjali Gupta	Mumbai	27
7	Karan Patel	Ahmedabad	31
NULL	NULL	NULL	NULL

Level 5 – Aggregations

SELECT COUNT(*) FROM customers;

COUNT(*)
8

SELECT AVG(price) FROM products;

Result Grid | Filter Rows:

AVG(price)
25850.000000

SELECT MAX(price) FROM products;

MAX(price)
80000.00

SELECT SUM(quantity) FROM sales;

Result Grid | Filter Row

SUM(quantity)
24

Level 6 – GROUP BY

SELECT category, COUNT(*) FROM products GROUP BY category;

category	COUNT(*)
Electronics	3
Clothing	2
Home Appliances	3

SELECT product_id, SUM(quantity) FROM sales GROUP BY product_id;

product_id	SUM(quantity)
101	2
102	3
104	8
105	3
103	6
106	1
108	1

SELECT category, AVG(price) FROM products GROUP BY category;

category	AVG(price)
Electronics	40666.666667
Clothing	1400.000000
Home Appliances	27333.333333

Level 7 – HAVING

```
SELECT category, AVG(price)
FROM products
GROUP BY category
HAVING AVG(price) > 10000;
```

Result Grid | Filter Rows:

category	AVG(price)
Electronics	40666.666667
Home Appliances	27333.333333

```
SELECT product_id, SUM(quantity)
FROM sales
GROUP BY product_id
HAVING SUM(quantity) > 3;
```

Result Grid | Filter Rows:

product_id	SUM(quantity)
104	8
103	6

```
SELECT category, COUNT(*)
FROM products
GROUP BY category
HAVING COUNT(*) > 2;
```

Result Grid | Filter Rows:

category	COUNT(*)
Electronics	3
Home Appliances	3

Business Questions

Most selling category

```
SELECT p.category, SUM(s.quantity) AS total_sales
FROM sales s
JOIN products p ON s.product_id = p.product_id
GROUP BY p.category
ORDER BY total_sales DESC
LIMIT 1;
```

Result Grid | Filter Rows:

category	total_sales
Electronics	11

Most sold product

```
SELECT product_id, SUM(quantity) AS total_sold
FROM sales
GROUP BY product_id
ORDER BY total_sold DESC;
```

Result Grid | Filter Rows:

product_id	total_sold
104	8
103	6
102	3
105	3
101	2
106	1
108	1

City with most customers

SELECT city, COUNT(*)
FROM customers
GROUP BY city
ORDER BY COUNT(*) DESC;

city	COUNT(*)
Delhi	2
Mumbai	2
Bangalore	1
Pune	1
Ahmedabad	1
NULL	1

Highest price product

SELECT * FROM products
ORDER BY price DESC
LIMIT 1;

product_id	product_name	category	price
101	Laptop	Electronics	80000.00
NULL	NULL	NULL	NULL